ECET 250E

Linear Circuits 1

Week 6 | Chapter 7: First-Order Circuits

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Fall 2026

Based on Fundamentals of Electric Circuits, 2026 Release — C.K. Alexander, M.N.O. Sadiku & S.J. Fredette (McGraw Hill)

Overview

- This chapter examines RC and LC circuits' reaction to switched sources.
- The circuits are referred to as first order circuits.
- Three special functions, the unit step, unit impulse, and unit ramp function are also introduced.
- Both source free and switched sources are examined.

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First Order Circuits

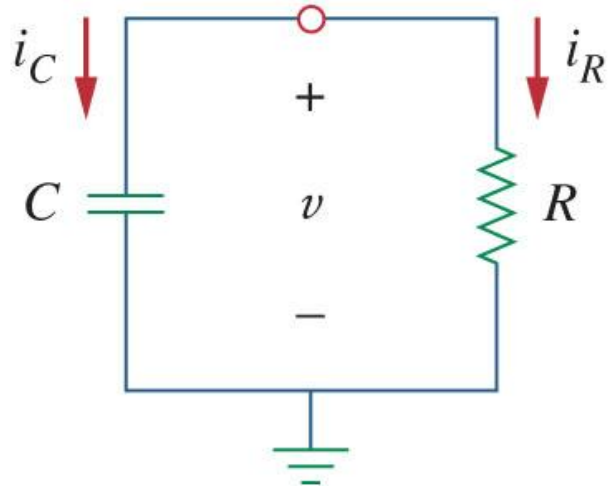
- A first order circuit is characterized by a first order differential equation.
- There are two types of first order circuits:
- Resistive capacitive, called **RC**.
- Resistive inductive, called **RL**.
- There are also two ways to excite the circuits:
- Initial conditions.
- Independent sources.

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Source Free RC Circuit

- **A source free** RC circuit occurs when its dc source is suddenly disconnected.
- The energy stored in the capacitor is released to the resistors.
- Consider a series combination of a resistor and a initially charged capacitor as shown:

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Source Free RC Circuit

- Since the capacitor was initially charged, we can assume at t=0 the initial voltages is:

$$v(0) = V_0$$

- Applying KCL at the top node:

$$i_C + i_R = 0$$

- Or

$$\frac{dv}{dt} + \frac{v}{RC} = 0$$

- This is a first order differential equation.

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Source Free RC Circuit 3

- Rearranging the equation and solving both sides yields(integrating both sides):

$$\ln v = -\frac{t}{RC} + \ln A$$

- Where A is the integration constant.
- Taking powers of e produces Taking powers of e produces.

$$v(t) = Ae^{-t/RC}$$

- With the initial conditions:

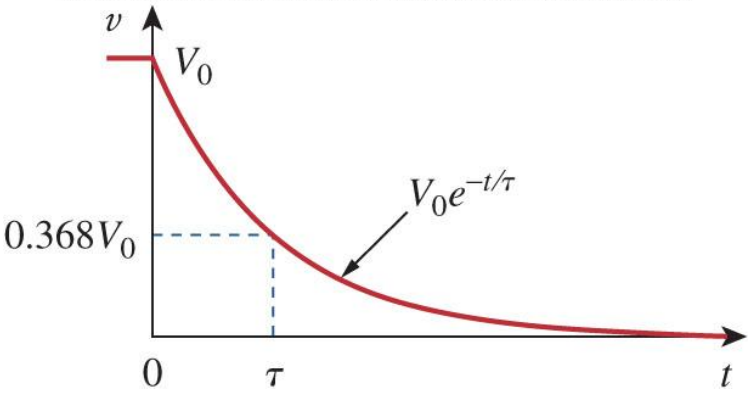
$$v(t) = V_0 e^{-t/RC}$$

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Natural Response

- The result shows that the voltage response of the RC circuit is an exponential decay of the initial voltage.
- Since this is the response of the circuit without any external applied voltage or current, the response is called the **natural response**.

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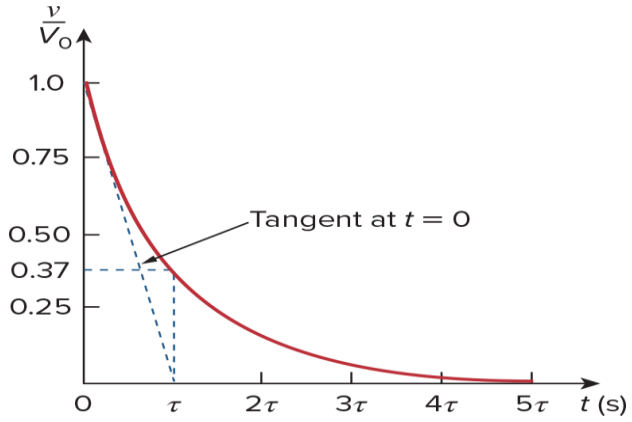


Time Constant

- The speed at which the voltage decays can be characterized by how long it takes the voltage to drop to $1/e$ of the initial voltage.
- This is called the time constant and is represented by τ.
- By selecting $1/e$ as the reference voltage: $\tau = RC$

The **time constant** of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value.[1]

- The voltage can thus be expressed as: $v(t) = V_0 e^{-t/\tau}$



Graphical determination of the time constant τ from the response curve.

TABLE 7.1
Values of $v(t)/V_0 = e^{-t/\tau}$.

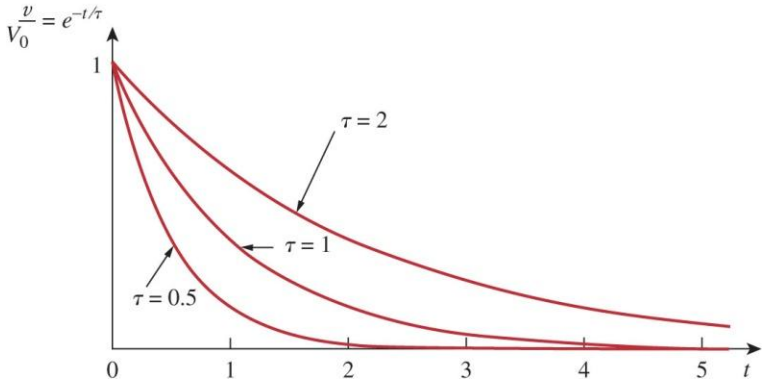
t	$v(t)/V_0$
τ	0.36788
2τ	0.13534
3τ	0.04979
4τ	0.01832
5τ	0.00674

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Time Constant 2

- After five time constants the voltage on the capacitor is less than one percent.
- After five time constants a capacitor is considered to be either fully discharged or charged.
- A circuit with a small time constant has a fast response and vice versa.

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Plot of v/V0=e^(−t/τ) for various values of the time constant.

RC Discharge

- With the voltage known, we can find the current:

$$i_R(t) = \frac{V_0}{R} e^{-t/\tau}$$

- The power dissipated in the resistor is:

$$p(t) = \frac{V_0^2}{R} e^{-2t/\tau}$$

- The energy absorbed by the resistor is:

$$w_R(t) = \frac{1}{2} C V_0^2 \left(1 - e^{-2t/\tau}\right)$$

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Source Free RC Circuit Summary

- The key to working with this type of situation is:
- Start with the initial voltage across the capacitor and the time constant.
- With these two items, the voltage as a function of time can be known.
- From the voltage, the current can be known by using the resistance and Ohm's law.
- The resistance of the circuit is often the Thevenin equivalent resistance.

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Example 7.1

In Fig. 7.5, let $v_C(0) = 15$ V. Find v_C, v_x, and i_x for $t > 0$.

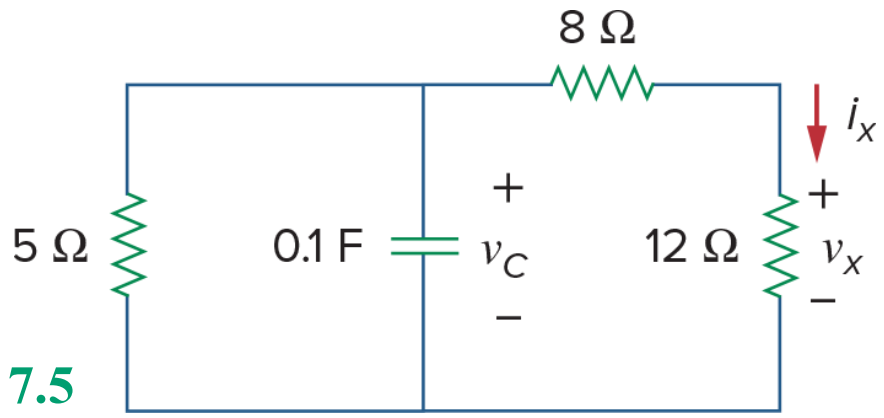


Figure 7.5
For Example 7.1.

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Example 7.2

The switch in the circuit in Fig. 7.8 has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

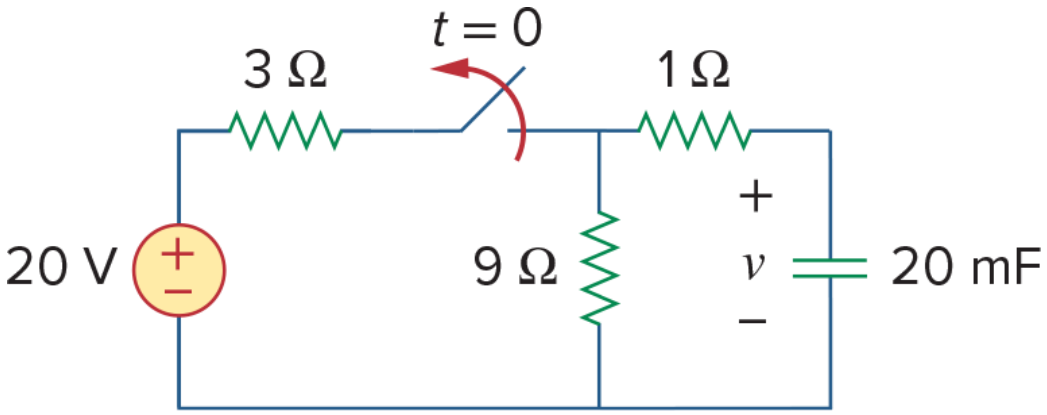


Figure 7.8
For Example 7.2.

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Source Free RL Circuit 1

- Now lets consider the series connection of a resistor and inductor.
- In this case, the value of interest is the current through the inductor.
- Since the current cannot change instantaneously, we can determine its value as a function of time.
- Once again, we will start with an initial current passing through the inductor.

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Source Free RL Circuit 2

- We will take the initial current to be:

$$i(0) = I_0$$

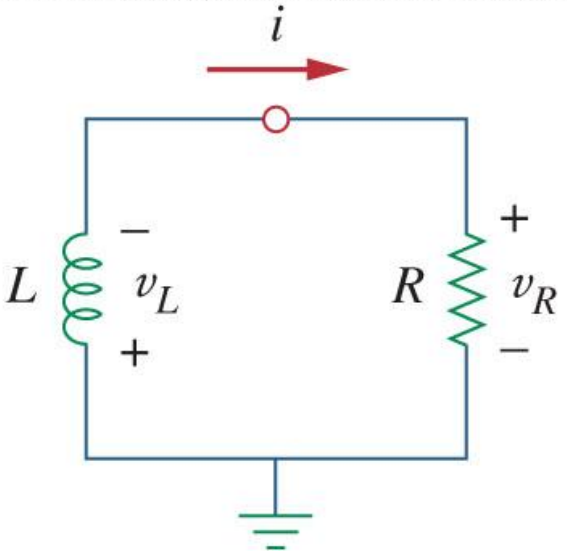
- Applying K V L around the loop:

$$v_L + v_R = 0$$

- Or:

$$L\frac{di}{dt} + Ri = 0$$

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Source Free RL Circuit 3

- After integration:

$$i(t) = I_0 e^{-Rt/L}$$

- Once again, the natural response is an exponentially decaying current.
- The time constant in this case is:

$$\tau = \frac{L}{R}$$

- The same principles as the RC circuit apply here.

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The switch in the circuit of Fig. 7.16 has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

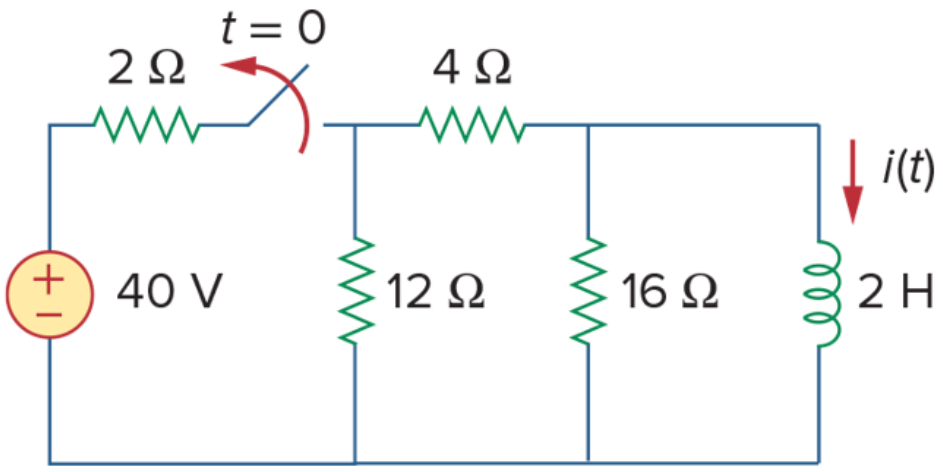
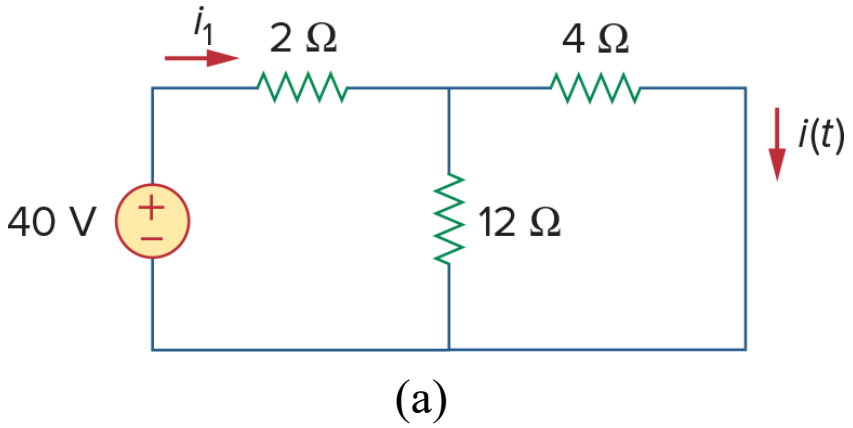
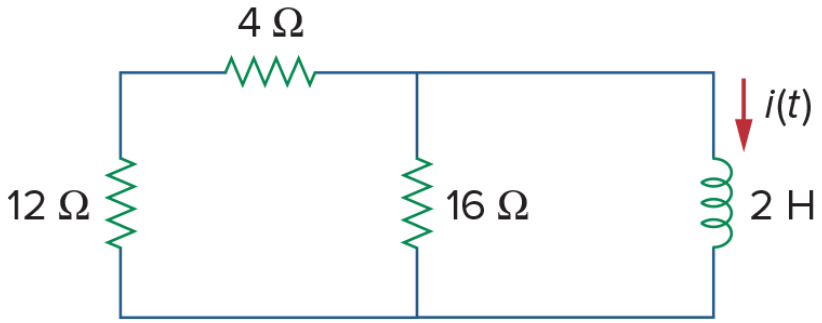


Figure 7.16 For **Example 7.4**.

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(a)



(b)

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Singularity Functions

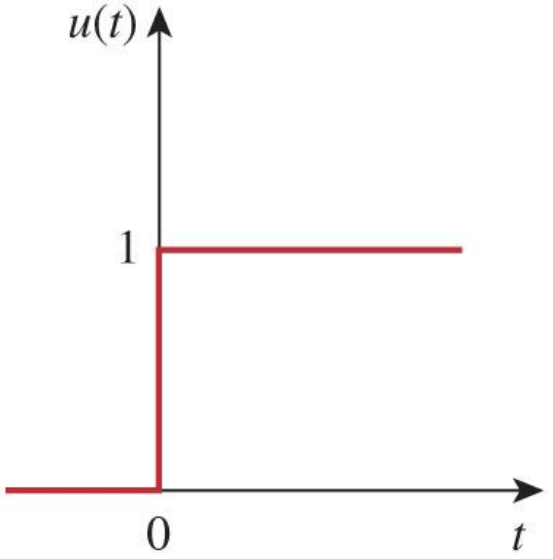
- Before we consider the response of a circuit to an external voltage, we need to cover some important mathematical functions.
- Singularity functions serve as good approximations to switching on or off a voltage.
- The three most common singularity functions are the unit step, unit impulse, and unit ramp.

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The Unit Step [1]

- A step function is one that maintains a constant value before a certain time and then changes to another constant afterwards.
- The prototypical form is zero before *t=0* and one afterwards.
- See the graph for an illustration.

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The Unit Step 2

- Mathematically, the unit step is expressed as:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t.

- The switching time may be shifted to $t=t_0$ by:

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

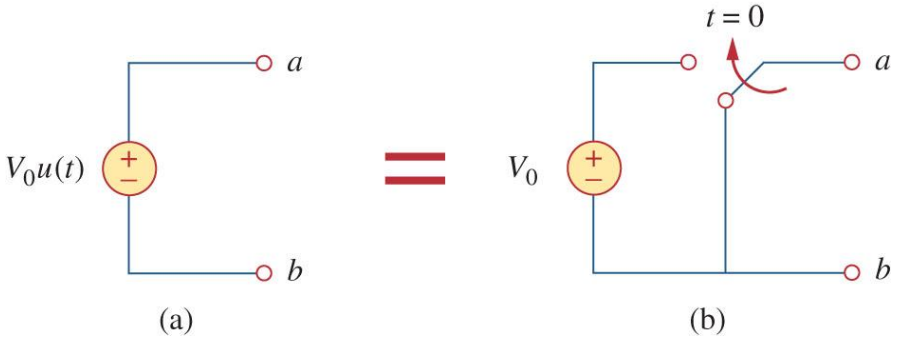
- Note that this results in a *delay* in the switch.
- The unit step function is written as *u(t)*

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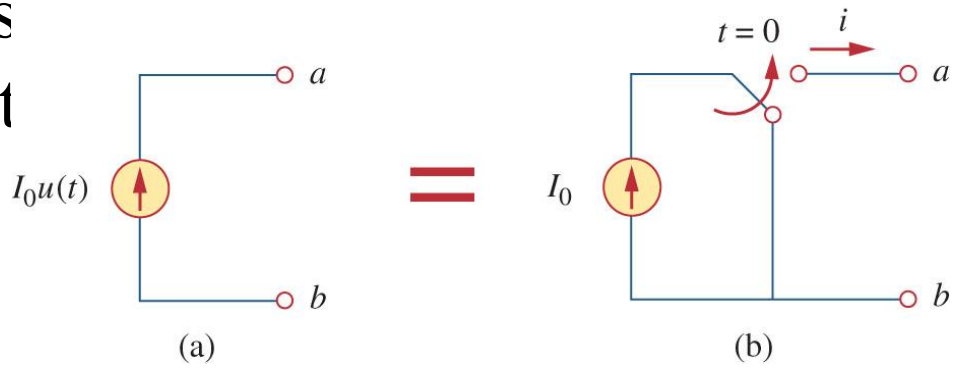
Equivalent Circuit

- The unit step function has an equivalent circuit to represent when it is used to switch on a source.
- The equivalent circuits for a voltage and current source are shown.

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The Unit Impulse Function

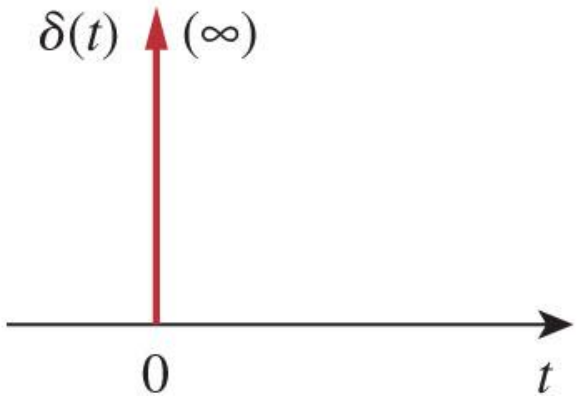
- The derivative of the unit step function is the unit impulse function.
- This is expressed as:

$$\delta(t) = \begin{cases} 0 & t < 0 \\ \text{Undefined} & t = 0 \\ 0 & t > 0 \end{cases}$$

The unit impulse function $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

- Voltages of this form can occur during switching operations.

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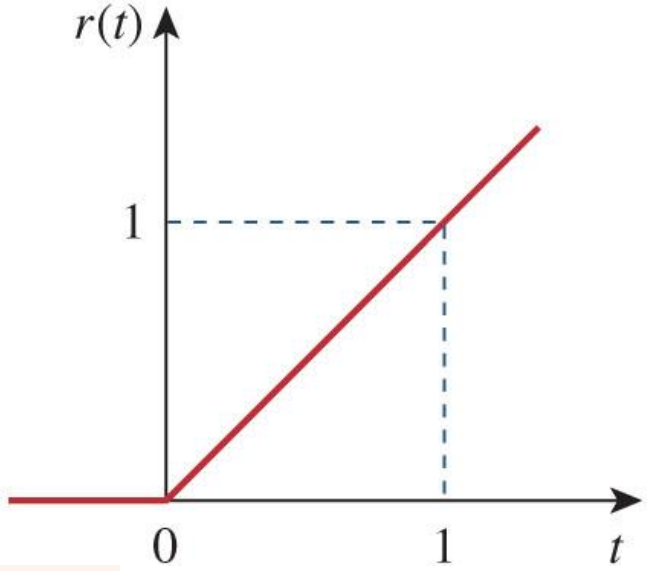
The unit impulse function—also known as the *delta* function

The Unit Ramp Function

- Integration of the unit step function results in the unit ramp function:

$$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$$

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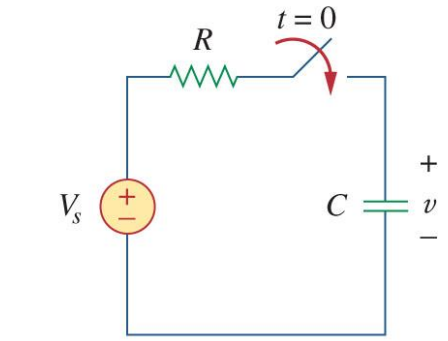
The unit ramp function is zero for negative values of t and has a unit slope for positive values of t.

- Much like the other functions, the onset of the ramp may be adjusted.

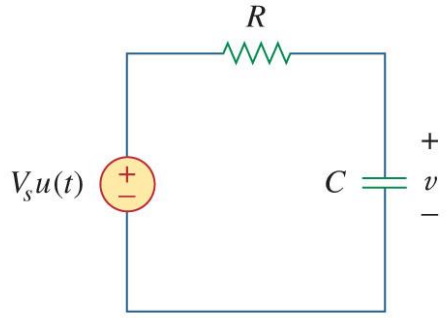
Step Response of RC Circuit 1

- When a DC source is suddenly applied to a RC circuit, the source can be modeled as a step function.
- The circuit response is known as the step response.
- Let's consider the circuit shown here.
- We can find the voltage on the capacitor as a function of time.

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(a)



(b)

Step Response of RC Circuit 2

- We assume an initial voltage of V_0 on the capacitor.
- Applying KCL:

$$\frac{dv}{dt}+\frac{v}{RC}=\frac{V_s}{RC}u(t)$$

- For $t>0$ this becomes:

$$\frac{dv}{dt}+\frac{v}{RC}=\frac{V_s}{RC}$$

- Integrating both sides and introducing initial conditions finally yields:

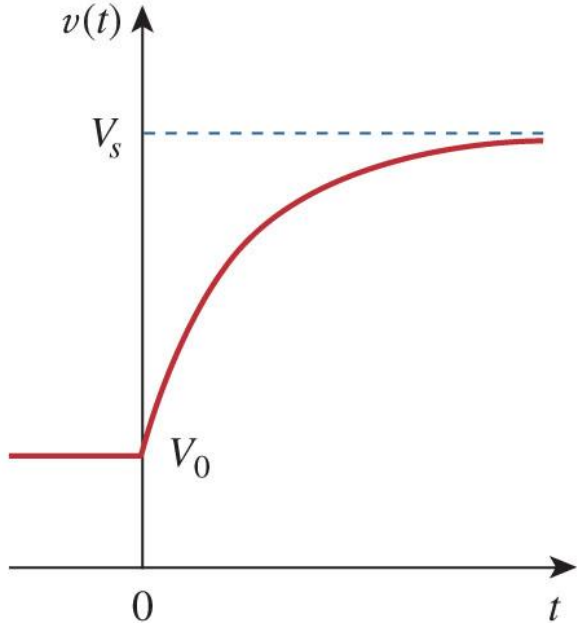
$$v(t)=\begin{cases} V_0, & t<0 \\ V_s+(V_0-V_s)e^{-t/\tau} & t>0 \end{cases}$$

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Step Response of RC Circuit 3

- This is known as the complete response, or total response.
- We can consider the response to be broken into two separate responses:
- The natural response of the capacitor or inductor due to the energy stored in it.
- The second part is the forced response.

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Forced Response

- The complete response can be written as:

$$v = v_n + v_f$$

- Where the nature response is:

$$v_n = V_0 e^{-t/\tau}$$

- And the forced response is:

$$v_f = V_s \left(1 - e^{-t/\tau}\right)$$

- Note that the eventual response of the circuit is to reach V_s after the natural response decays to zero.

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Another Perspective

- Another way to look at the response is to break it up into the transient response and the steady state response:

$$v = v_t + v_{ss}$$

- Where the transient is:

$$v_t = \left(V_0 - V_s\right) e^{-t/\tau}$$

- And the steady state is:

$$v_{ss} = V_s$$

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Example 7.10

The switch in Fig. 7.43 has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ and 4 s.

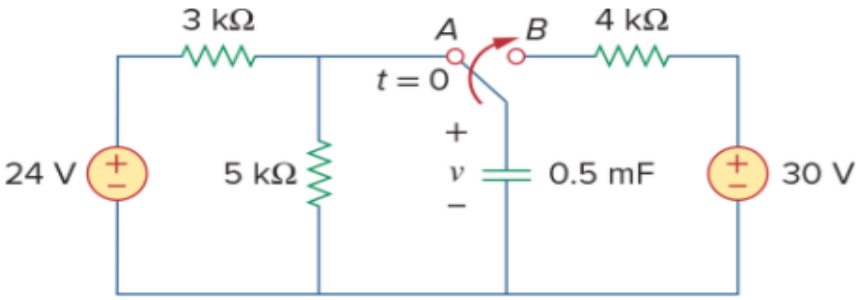


Figure 7.43
For Example 7.10.

Solution:

For $t < 0$, the switch is at position A. The capacitor acts like an open circuit to dc, but v is the same as the voltage across the 5-kΩ resistor. Hence, the voltage across the capacitor just before $t = 0$ is obtained by voltage division as

$$v(0^-) = \frac{5}{5+3}(24) = 15 \text{ V}$$

Using the fact that the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = v(0^+) = 15 \text{ V}$$

For $t > 0$, the switch is in position B. The Thevenin resistance connected to the capacitor is $R_{\text{Th}} = 4$ kΩ, and the time constant is

$$\tau = R_{\text{Th}} C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s}$$

Since the capacitor acts like an open circuit to dc at steady state, $v(\infty) = 30$ V. Thus,

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 30 + (15 - 30)e^{-t/2} = \left(30 - 15e^{-0.5t}\right) \text{ V} \end{aligned}$$

At $t = 1$,

$$v(1) = 30 - 15e^{-0.5} = 20.9 \text{ V}$$

At $t = 4$,

$$v(4) = 30 - 15e^{-2} = 27.97 \text{ V}$$

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Example 7.11

In Fig. 7.45, the switch has been closed for a long time and is opened at $t = 0$. Find i and v for all time.

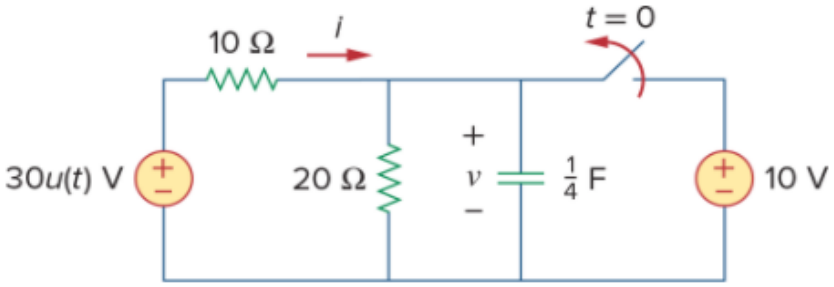


Figure 7.45
For Example 7.11.

Solution:

The resistor current i can be discontinuous at $t = 0$, while the capacitor voltage v cannot. Hence, it is always better to find v and then obtain i from v.

By definition of the unit step function,

$$30u(t) = \begin{cases} 0, & t < 0 \\ 30, & t > 0 \end{cases}$$

For $t < 0$, the switch is closed and $30u(t) = 0$, so that the $30u(t)$ voltage source is replaced by a short circuit and should be regarded as contributing nothing to v. Since the switch has been closed for a long time, the capacitor voltage has reached steady state and the capacitor acts like an open circuit. Hence, the circuit becomes that shown in Fig. 7.46(a) for $t < 0$. From this circuit we obtain

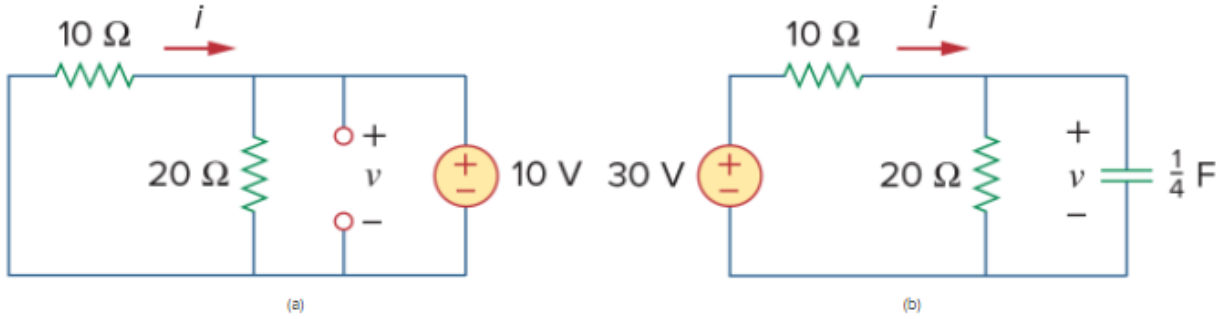


Figure 7.46
Solution of Example 7.11: (a) for $t < 0$, (b) for $t > 0$.

$$v = 10 \text{ V}, \quad i = -\frac{v}{10} = -1 \text{ A}$$

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Since the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = 10 \text{ V}$$

For $t > 0$, the switch is opened and the 10-V voltage source is disconnected from the circuit. The $30u(t)$ voltage source is now operative, so the circuit becomes that shown in Fig. 7.46(b). After a long time, the circuit reaches steady state and the capacitor acts like an open circuit again. We obtain $v(\infty)$ by using voltage division, writing

$$v(\infty) = \frac{20}{20 + 10}(30) = 20 \text{ V}$$

The Thevenin resistance at the capacitor terminals is

$$R_{\text{Th}} = 10 \| 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega$$

and the time constant is

$$\tau = R_{\text{Th}} C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3} \text{ s}$$

Thus,

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = \left(20 - 10e^{-0.6t}\right) \text{ V} \end{aligned}$$

To obtain i, we notice from Fig. 7.46(b) that i is the sum of the currents through the 20-Ω resistor and the capacitor; that is,

$$\begin{aligned} i &= \frac{v}{20} + C\frac{dv}{dt} \\ &= 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = \left(1 + e^{-0.6t}\right) \text{ A} \end{aligned}$$

Notice from Fig. 7.46(b) that $v + 10i = 30$ is satisfied, as expected. Hence,

$$v = \begin{cases} 10 \text{ V}, & t < 0 \\ \left(20 - 10e^{-0.6t}\right) \text{ V}, & t \geq 0 \end{cases}$$

$$i = \begin{cases} -1 \text{ A}, & t < 0 \\ \left(1 + e^{-0.6t}\right) \text{ A}, & t > 0 \end{cases}$$

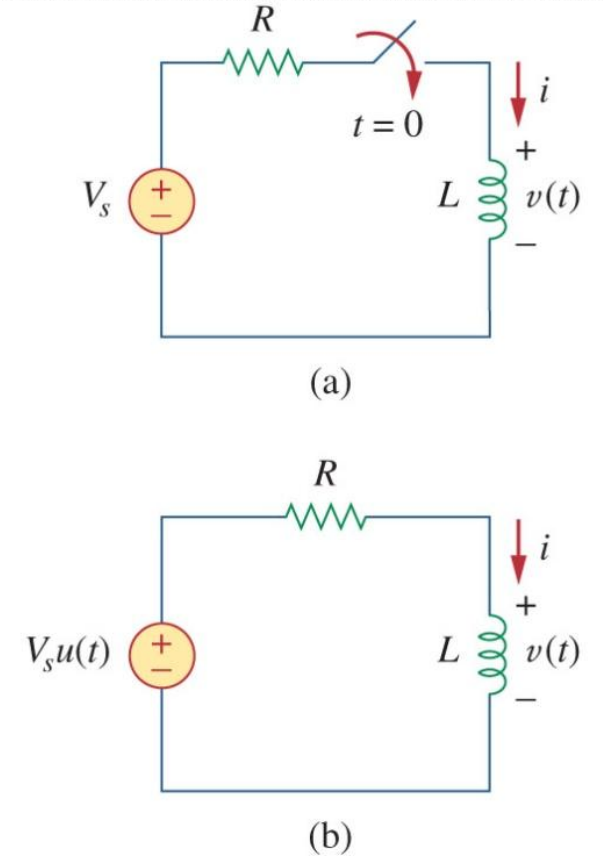
Notice that the capacitor voltage is continuous while the resistor current is not.

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Step Response of RL Circuit [1]

- Now we can look at the step response of a RL circuit.
- We will use the transient and steady state response approach.
- We know that the transient response will be an exponential:

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$$i_t = Ae^{-t/\tau}$$

Step Response of RL Circuit 2

- After a sufficiently long time, the current will reach he steady state:

$$i_{ss} = \frac{V_s}{R}$$

- This yields an overall response of:

$$i = Ae^{-t/\tau} + \frac{V_s}{R}$$

- To determine the value of A we need to keep in mind that the current cannot change instantaneously.

$$i\left(0^+\right) = i\left(0^-\right) = I_0$$

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Step Response of RL Circuit 3

- Thus we can use the *t=0* time to establish *A*.

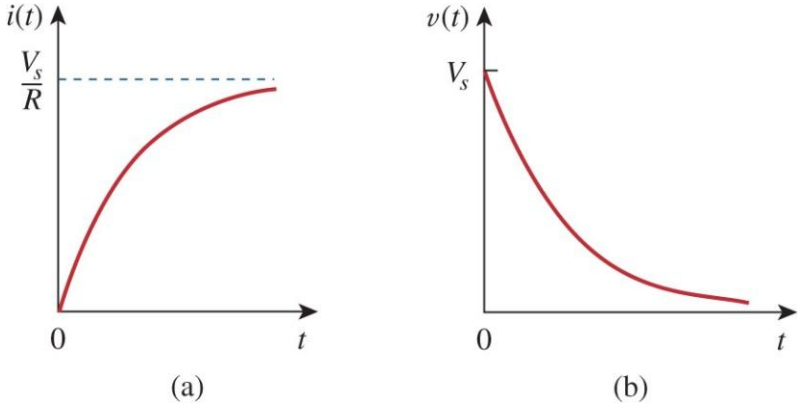
$$A = I_0 - \frac{V_s}{R}$$

- The complete response of the circuit is thus:

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

- Without an initial current, the circuit response is shown here.

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Example 7.12

Find $i(t)$ in the circuit of Fig. 7.51 for $t > 0$. Assume that the switch has been closed for a long time.

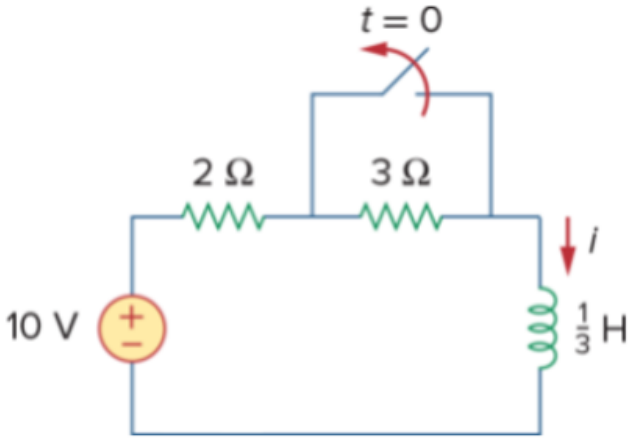


Figure 7.51
For Example 7.12.

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